Controller RTL Verification

VCS + Verdi | Original RTL, high-contrast stress fixture

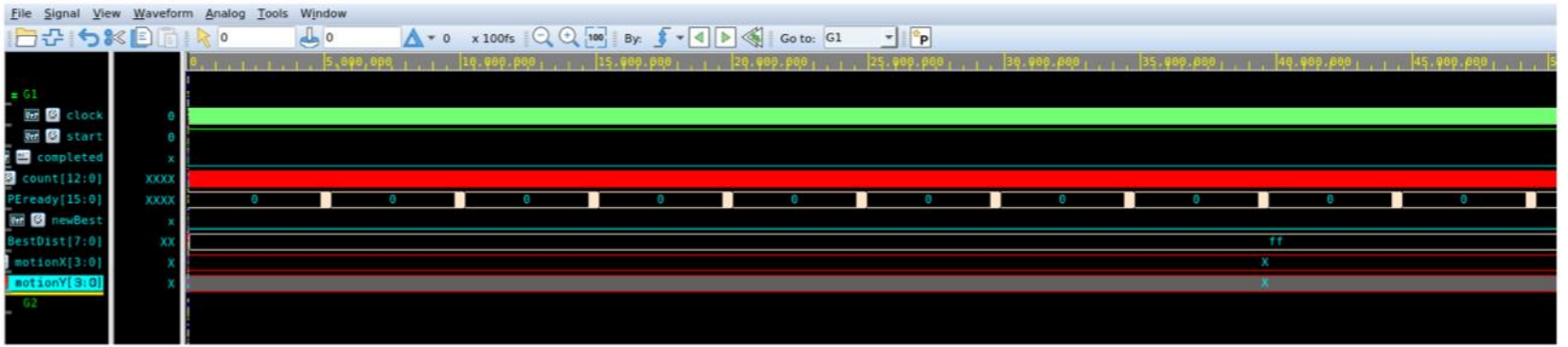
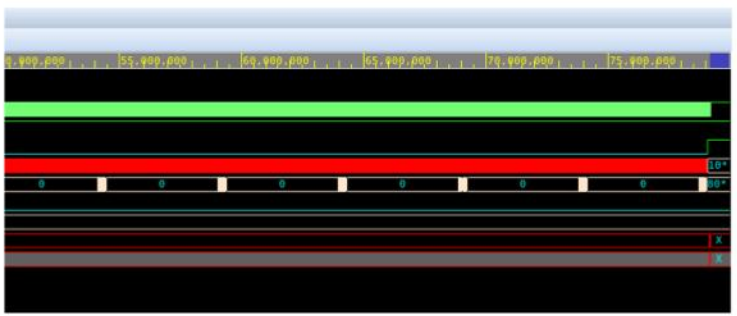


Figure 1. Clock, PEready, count, newBest and output registers across the test.

Figure 2. End of the simulation window



Original 8-bit result

BestDist = FF

motionX, motionY = X, X

The saturated comparator never chooses a valid candidate in this stress test.

16-bit SAD Verification

VCS + Verdi | Separate precision-corrected RTL and directed stimulus

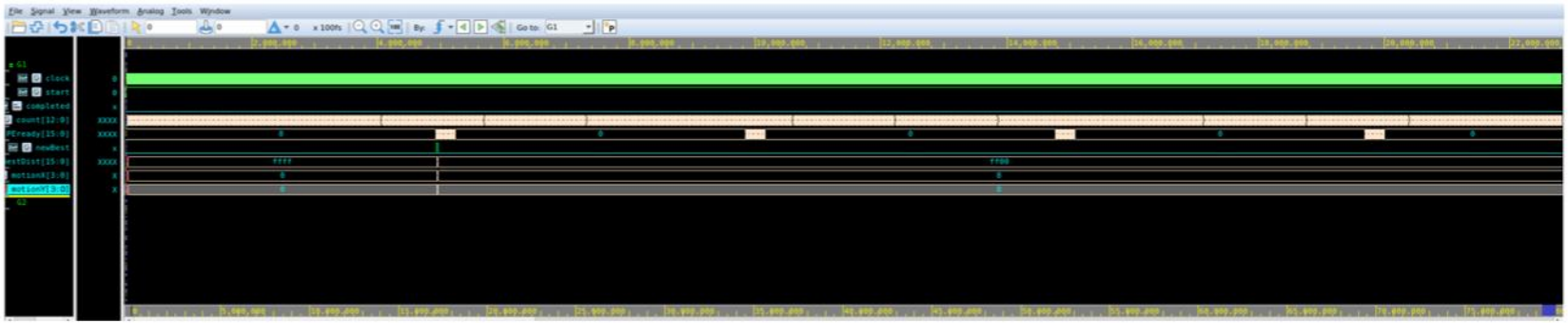


Figure 3. newBest selects FF00 and the encoded vector (8,8) for the all-FF stress case.

BestDist is input-dependent: these are three different stimulus cases.

Team perfect match	**0000**
Single-pixel perturbation	**0003**
All-00 reference / all-FF search	**FF00**

Exact match: SAD = 0x0000 | Maximum contrast: SAD = 0xFF00

14 nm Physical Design Checkpoint

Fusion Compiler + PrimeTime | 520 MHz | Routed V5 results

Physical implementation

Measured metric	V5 result
Routed cell area	2,781.30 μm²
Mapped cells	6,310
Sequential cells	449
Memory macros	0
Typical power	0.729 mW

PrimeTime: extracted RC

Corner	Setup WNS	Hold WNS
FAST	+0.52 ns	-0.14 ns
TYPICAL	+0.30 ns	-0.12 ns
SLOW	-0.06 ns	-0.09 ns

Setup: 17 SLOW failures
Hold: 402 FAST / 120 TYP / 33 SLOW

ICV DRC (routed V5): 18 geometry violations measured.

520 MHz SDC: 1.923 ns period | 0.5 ns I/O delay | 0.2 ns uncertainty